Formally Verified PCP Constructions

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Chapter 1

BLR for general finite fields

1.1 BLR over General Finite Fields

Throughout this section, let $q = p^s$ for a prime p and $s \in \mathbb{N}$ with $s \geq 1$. All vectors are in $\mathbb{F}_q^n$, and all expectations are uniform over the indicated finite sets. Let $\omega_p = e^{2\pi i/p}$.

Definition 1.1.1 (Field trace)**.** *The field trace* $\mathrm{Tr} : \mathbb{F}_q \to \mathbb{F}_p$ *is*

$$\mathrm{Tr}(z) := \sum_{i=0}^{s-1} z^{p^i}.$$

Definition 1.1.2 (Base additive character)**.** *The base additive character* $\psi : \mathbb{F}_q \to \mathbb{C}$ *is*

$$\psi(z) := \omega_p^{\mathrm{Tr}(z)}.$$

Definition 1.1.3 (Additive characters)**.** *For* $\alpha \in \mathbb{F}_q^n$*, define the additive character* $\chi_\alpha : \mathbb{F}_q^n \to \mathbb{C}$ *by*

$$\chi_\alpha(x) := \psi(\langle \alpha, x \rangle) = \omega_p^{\mathrm{Tr}(\langle \alpha, x \rangle)},$$

where $\langle \alpha, x \rangle = \sum_{i=1}^n \alpha_i x_i \in \mathbb{F}_q$*.*

Definition 1.1.4 (Expectation)**.** *For a function* $h : \mathbb{F}_q^n \to \mathbb{C}$*, define its uniform expectation by*

$$\mathbb{E}_{x \in \mathbb{F}_q^n}[h(x)] := \frac{1}{q^n} \sum_{x \in \mathbb{F}_q^n} h(x).$$

Definition 1.1.5 (Inner product)**.** *For functions* $h_1, h_2 : \mathbb{F}_q^n \to \mathbb{C}$*, define*

$$\langle h_1, h_2 \rangle := \mathbb{E}_{x \in \mathbb{F}_q^n}\left[h_1(x)\overline{h_2(x)}\right].$$

Definition 1.1.6 (L_2 norm)**.** *For a function* $h : \mathbb{F}_q^n \to \mathbb{C}$*, define its* L_2 *norm by*

$$\|h\|_2 := \left(\mathbb{E}_{x \in \mathbb{F}_q^n}\left[|h(x)|^2\right]\right)^{1/2} = \sqrt{\langle h, h \rangle}.$$

Definition 1.1.7 (Fourier coefficient)**.** *For* $h : \mathbb{F}_q^n \to \mathbb{C}$ *and* $\alpha \in \mathbb{F}_q^n$*, define*

$$\hat{h}(\alpha) := \langle h, \chi_\alpha \rangle = \mathbb{E}_{x \in \mathbb{F}_q^n}\left[h(x)\overline{\chi_\alpha(x)}\right].$$

Lemma 1.1.8 (Characters are orthonormal)**.** *For every* $\alpha, \beta \in \mathbb{F}_q^n$*,*

$$\langle \chi_\alpha, \chi_\beta \rangle = \begin{cases} 1 & \text{if } \alpha = \beta, \\ 0 & \text{if } \alpha \neq \beta. \end{cases}$$

Proof. First observe that for every $\alpha, \beta \in \mathbb{F}_q^n$ and every $x \in \mathbb{F}_q^n$,

$$\chi_\alpha(x)\overline{\chi_\beta(x)} = \omega_p^{\mathrm{Tr}(\langle \alpha, x\rangle)}\omega_p^{-\mathrm{Tr}(\langle \beta, x\rangle)} = \omega_p^{\mathrm{Tr}(\langle \alpha-\beta, x\rangle)} = \chi_{\alpha-\beta}(x),$$

where we used the $\mathbb{F}_p$-linearity of the trace map.

If $\alpha = \beta$, then $\alpha - \beta = 0$, so $\chi_{\alpha-\beta}(x) = 1$ for every $x \in \mathbb{F}_q^n$. Hence

$$\mathbb{E}_{x \in \mathbb{F}_q^n}\left[\chi_\alpha(x)\overline{\chi_\beta(x)}\right] = \mathbb{E}_{x \in \mathbb{F}_q^n}[1] = 1.$$

Now suppose $\alpha \neq \beta$. Let $\gamma \coloneqq \alpha - \beta$. Then $\gamma \neq 0$, so there exists an index j such that $\gamma_j \neq 0$. Fix all coordinates of x except x_j. For the fixed remaining coordinates, the map

$$x_j \mapsto \langle \gamma, x \rangle$$

has the form

$$x_j \mapsto \gamma_j x_j + c$$

for some constant $c \in \mathbb{F}_q$. Since $\gamma_j \neq 0$, multiplication by γ_j is a bijection on $\mathbb{F}_q$, and therefore the affine map $x_j \mapsto \gamma_j x_j + c$ is also a bijection on $\mathbb{F}_q$. Thus, as x_j ranges uniformly over $\mathbb{F}_q$, the value $\langle \gamma, x\rangle$ also ranges uniformly over $\mathbb{F}_q$. Averaging first over x_j and then over the remaining coordinates gives

$$\mathbb{E}_{x \in \mathbb{F}_q^n}\left[\chi_\gamma(x)\right] = \mathbb{E}_{t \in \mathbb{F}_q}\left[\omega_p^{\mathrm{Tr}(t)}\right].$$

It remains to show that this last average is zero. The trace map $\mathrm{Tr} : \mathbb{F}_q \to \mathbb{F}_p$ is a nonzero $\mathbb{F}_p$-linear map, so every fiber of Tr has the same cardinality. Since $|\mathbb{F}_q| = q$ and $|\mathbb{F}_p| = p$, each fiber has cardinality q/p. Therefore

$$\begin{aligned} \mathbb{E}_{t \in \mathbb{F}_q}\left[\omega_p^{\mathrm{Tr}(t)}\right] &= \frac{1}{q}\sum_{t \in \mathbb{F}_q}\omega_p^{\mathrm{Tr}(t)} \\ &= \frac{1}{q}\sum_{r \in \mathbb{F}_p}\sum_{\substack{t \in \mathbb{F}_q \\ \mathrm{Tr}(t)=r}}\omega_p^r \\ &= \frac{1}{q}\sum_{r \in \mathbb{F}_p}\frac{q}{p}\omega_p^r \\ &= \frac{1}{p}\sum_{r \in \mathbb{F}_p}\omega_p^r \\ &= 0, \end{aligned}$$

because the sum of all p-th roots of unity is zero. Hence, if $\alpha \neq \beta$, then

$$\mathbb{E}_{x \in \mathbb{F}_q^n}\left[\chi_\alpha(x)\overline{\chi_\beta(x)}\right] = 0.$$

Combining the two cases proves the stated orthogonality relation. The equivalent formulation follows by taking $\beta = 0$. $\square$

Lemma 1.1.9 (Fourier inversion)**.** *For every function $h : \mathbb{F}_q^n \to \mathbb{C}$ we have*

$$h(x) = \sum_{\alpha \in \mathbb{F}_q^n} \hat{h}(\alpha)\chi_\alpha(x) \quad \textit{for every } x \in \mathbb{F}_q^n.$$

Proof. By Theorem 1.1.8, the additive characters

$$\{\chi_\alpha : \alpha \in \mathbb{F}_q^n\}$$

are orthonormal with respect to the inner product

$$\langle h_1, h_2 \rangle = \mathbb{E}_{x \in \mathbb{F}_q^n}\left[h_1(x)\overline{h_2(x)}\right].$$

Moreover, these characters span the vector space of all complex-valued functions on $\mathbb{F}_q^n$. Indeed, there are exactly q^n characters, and the space of functions $\mathbb{F}_q^n \to \mathbb{C}$ has dimension q^n. Since the characters are nonzero and pairwise orthogonal, they are linearly independent. Hence they form an orthonormal basis.

Therefore every function $h : \mathbb{F}_q^n \to \mathbb{C}$ has a unique expansion in this basis:

$$h = \sum_{\alpha \in \mathbb{F}_q^n} c_\alpha \chi_\alpha$$

for some coefficients $c_\alpha \in \mathbb{C}$. Taking the inner product of both sides with χ_β, we obtain

$$\langle h, \chi_\beta \rangle = \sum_{\alpha \in \mathbb{F}_q^n} c_\alpha \langle \chi_\alpha, \chi_\beta \rangle.$$

By orthonormality, all terms in the sum vanish except the term $\alpha = \beta$, and $\langle \chi_\beta, \chi_\beta \rangle = 1$. Hence

$$\langle h, \chi_\beta \rangle = c_\beta.$$

By definition of the Fourier coefficient,

$$\hat{h}(\beta) = \langle h, \chi_\beta \rangle.$$

Thus $c_\beta = \hat{h}(\beta)$ for every $\beta \in \mathbb{F}_q^n$. Therefore

$$h = \sum_{\alpha \in \mathbb{F}_q^n} \hat{h}(\alpha)\chi_\alpha.$$

Evaluating this identity at $x \in \mathbb{F}_q^n$ gives the Fourier inversion formula

$$h(x) = \sum_{\alpha \in \mathbb{F}_q^n} \hat{h}(\alpha)\chi_\alpha(x).$$

□

Lemma 1.1.10 (Plancherel identity)**.** *For every pair of functions $f, g : \mathbb{F}_q^n \to \mathbb{C}$,*

$$\sum_{\alpha \in \mathbb{F}_q^n} \hat{f}(\alpha)\overline{\hat{g}(\alpha)} = \langle f, g \rangle.$$

Proof. By Theorem 1.1.9,

$$f = \sum_{\alpha \in \mathbb{F}_q^n} \hat{f}(\alpha)\chi_\alpha.$$

Taking the inner product with g and using linearity in the first argument gives

$$\langle f, g\rangle = \sum_{\alpha \in \mathbb{F}_q^n} \hat{f}(\alpha)\langle \chi_\alpha, g\rangle.$$

Since

$$\langle \chi_\alpha, g\rangle = \overline{\langle g, \chi_\alpha\rangle} = \overline{\hat{g}(\alpha)},$$

we obtain

$$\langle f, g\rangle = \sum_{\alpha \in \mathbb{F}_q^n} \hat{f}(\alpha)\overline{\hat{g}(\alpha)}.$$

□

Lemma 1.1.11 (Parseval identity). *For every function $h : \mathbb{F}_q^n \to \mathbb{C}$,*

$$\sum_{\alpha \in \mathbb{F}_q^n} \left|\hat{h}(\alpha)\right|^2 = \|h\|_2^2.$$

Proof. Apply Theorem 1.1.10 with $f = g = h$:

$$\sum_{\alpha \in \mathbb{F}_q^n} \hat{h}(\alpha)\overline{\hat{h}(\alpha)} = \langle h, h\rangle.$$

The left-hand side is

$$\sum_{\alpha \in \mathbb{F}_q^n} \left|\hat{h}(\alpha)\right|^2,$$

and by the definitions of the inner product and the L_2 norm, the right-hand side is

$$\langle h, h\rangle = \mathbb{E}_{x \in \mathbb{F}_q^n}\left[|h(x)|^2\right] = \|h\|_2^2.$$

□

Definition 1.1.12 (Nonzero field elements). *Write*

$$\mathbb{F}_q^\times := \{c \in \mathbb{F}_q : c \neq 0\}.$$

Definition 1.1.13 (Fourier expansion of a finite-field-valued function). *Let $f : \mathbb{F}_q^n \to \mathbb{F}_q$. For each $c \in \mathbb{F}_q^\times$, define the c-phase lift*

$$\varphi_c(x) := \omega_p^{\mathrm{Tr}(cf(x))}.$$

The Fourier expansion of f is the set $\{\varphi_c\}_{c \in \mathbb{F}_q^\times}$.

Lemma 1.1.14 (Character-sum indicator). *For every $u, v \in \mathbb{F}_q$,*

$$\mathbb{1}[u = v] = \frac{1}{q}\sum_{c \in \mathbb{F}_q} \omega_p^{\mathrm{Tr}(c(u-v))}.$$

Proof. For each $z \in \mathbb{F}_q$ we define

$$\psi(z) := \omega_p^{\mathrm{Tr}(z)}$$

be the standard additive character of $\mathbb{F}_q$. We prove that, for every $t \in \mathbb{F}_q$,

$$\mathbb{1}[t = 0] = \frac{1}{q} \sum_{c \in \mathbb{F}_q} \psi(ct).$$

The desired statement follows by taking $t = u - v$.

If $t = 0$, then $ct = 0$ for every $c \in \mathbb{F}_q$, so

$$\frac{1}{q} \sum_{c \in \mathbb{F}_q} \psi(ct) = \frac{1}{q} \sum_{c \in \mathbb{F}_q} 1 = 1.$$

Now suppose $t \neq 0$. Then multiplication by t is a bijection $\mathbb{F}_q \to \mathbb{F}_q$, so

$$\sum_{c \in \mathbb{F}_q} \psi(ct) = \sum_{z \in \mathbb{F}_q} \psi(z).$$

We claim that the latter sum is 0. Let

$$S := \sum_{z \in \mathbb{F}_q} \psi(z).$$

Since $\mathrm{Tr} : \mathbb{F}_q \to \mathbb{F}_p$ is a nonzero $\mathbb{F}_p$-linear map, it is surjective. Hence there exists $a \in \mathbb{F}_q$ such that $\mathrm{Tr}(a) = 1$. Using the bijection $z \mapsto z + a$, we get

$$S = \sum_{z \in \mathbb{F}_q} \psi(z + a) = \sum_{z \in \mathbb{F}_q} \omega_p^{\mathrm{Tr}(z+a)} = \sum_{z \in \mathbb{F}_q} \omega_p^{\mathrm{Tr}(z)} \omega_p^{\mathrm{Tr}(a)} = \omega_p S.$$

Since $\omega_p \neq 1$, this implies $S = 0$. Therefore, when $t \neq 0$,

$$\frac{1}{q} \sum_{c \in \mathbb{F}_q} \psi(ct) = 0.$$

Combining the two cases gives

$$\mathbb{1}[t = 0] = \frac{1}{q} \sum_{c \in \mathbb{F}_q} \omega_p^{\mathrm{Tr}(ct)}.$$

Substituting $t = u - v$, we obtain

$$\mathbb{1}[u = v] = \frac{1}{q} \sum_{c \in \mathbb{F}_q} \omega_p^{\mathrm{Tr}(c(u-v))}.$$

$\square$

Definition 1.1.15 (Relative Hamming distance)**.** *For $f, g : \mathbb{F}_q^n \to \mathbb{F}_q$, define*

$$\delta(f, g) := \Pr_{x \in \mathbb{F}_q^n} [f(x) \neq g(x)].$$

Definition 1.1.16 (Linear functions)**.** *For $\alpha \in \mathbb{F}_q^n$, define the linear scalar function $\ell_\alpha : \mathbb{F}_q^n \to \mathbb{F}_q$ by*

$$\ell_\alpha(x) := \langle \alpha, x \rangle.$$

A function $h : \mathbb{F}_q^n \to \mathbb{F}_q$ is linear if $h = \ell_\alpha$ for some $\alpha \in \mathbb{F}_q^n$. The finite set of all linear scalar functions is

$$\mathcal{LIN} := \left\{ \ell_\alpha \mid \alpha \in \mathbb{F}_q^n \right\}.$$

Definition 1.1.17 (Distance to linearity)**.** *The distance from $f : \mathbb{F}_q^n \to \mathbb{F}_q$ to linearity is*

$$\delta(f, \mathcal{LIN}) := \min_{g \in \mathcal{LIN}} \delta(f, g).$$

Lemma 1.1.18 (Well-separatedness of $\mathcal{LIN}$)**.** *For any distinct $f, g \in \mathcal{LIN}$, one has $\delta(f,g) = 1 - 1/|\mathbb{F}|$.*

Proof. Let $a, b \in \mathbb{F}^n$ be the unique vectors such that $f(x) = \langle a, x \rangle$ and $g(x) = \langle b, x \rangle$ for each x. Let $c = a - b$, which must be a non-zero vector. In particular, there exists $i \in [n]$ such that $c_i \neq 0$. Take any permutation $\sigma : [n] \to [n]$ such that $\sigma(i) = n$. Then

$$\begin{aligned}
\delta(f,g) &= 1 - \Pr_{x \sim \mathbb{F}^n}[\langle c, x \rangle = 0] \\
&= 1 - \frac{1}{|\mathbb{F}|^{n-1}} \sum_{x_1, \ldots, x_{n-1} \in \mathbb{F}} \frac{1}{|\mathbb{F}|} \sum_{x_n \in \mathbb{F}} \mathbb{1}[c_i \cdot x_n = - \sum_{j \in [n] \setminus i} c_j \cdot x_{\sigma(j)}] \\
&= 1 - \frac{1}{|\mathbb{F}|^{n-1}} \sum_{x_1, \ldots, x_{n-1} \in \mathbb{F}} \frac{1}{|\mathbb{F}|} \sum_{x_n \in \mathbb{F}} \mathbb{1}[x_n = -c_i^{-1} \cdot \sum_{j \in [n] \setminus i} c_j \cdot x_{\sigma(j)}]
\end{aligned}$$

where the last equality follows because $c_i \neq 0$. Now note that $\mathbb{1}[x_n = -c_i^{-1} \cdot \sum_{j \in [n] \setminus i} c_j \cdot x_{\sigma(j)}]$ evaluates to 1 for exactly one value of x_n. Thus we continue from above and get

$$\delta(f,g) = 1 - \frac{1}{|\mathbb{F}|^{n-1}} \sum_{x_1, \ldots, x_{n-1} \in \mathbb{F}} \frac{1}{|\mathbb{F}|} = 1 - \frac{1}{|\mathbb{F}|}.$$

□

Lemma 1.1.19 (Distance formula)**.** *Let $f, g : \mathbb{F}_q^n \to \mathbb{F}_q$. Let $\{\varphi_c\}_{c \in \mathbb{F}_q^\times}$ and $\{\gamma_c\}_{c \in \mathbb{F}_q^\times}$ be their Fourier expansions. Then*

$$\delta(f,g) = 1 - \frac{1}{q} \left(1 + \sum_{c \in \mathbb{F}_q^\times} \langle \varphi_c, \gamma_c \rangle \right).$$

Proof. By Theorem 1.1.14,

$$\Pr_x[f(x) = g(x)] = \mathbb{E}_x \left[\frac{1}{q} \sum_{c \in \mathbb{F}_q} \omega_p^{\mathrm{Tr}(c(f(x) - g(x)))} \right].$$

The term $c = 0$ contributes $1/q$. For $c \neq 0$,

$$\omega_p^{\mathrm{Tr}(c(f(x)-g(x)))} = \varphi_c(x) \overline{\gamma_c(x)}.$$

Therefore

$$\Pr_x[f(x) = g(x)] = \frac{1}{q} \left(1 + \sum_{c \in \mathbb{F}_q^\times} \langle \varphi_c, \gamma_c \rangle \right),$$

which gives the first formula after using $\delta(f,g) = 1 - \Pr_x[f(x) = g(x)]$. □

Lemma 1.1.20 (Fourier expansion of a linear function)**.** *Fix $\alpha \in \mathbb{F}_q^n$ and let $\{\lambda_c\}_{c\in\mathbb{F}_q^\times}$ be the Fourier expansion of ℓ_α. Then*

$$\lambda_c = \chi_{c\alpha}.$$

Proof. For every $x \in \mathbb{F}_q^n$,

$$\lambda_c(x) = \omega_p^{\mathrm{Tr}(c\langle\alpha,x\rangle)} = \omega_p^{\mathrm{Tr}(\langle c\alpha,x\rangle)} = \chi_{c\alpha}(x).$$

□

Definition 1.1.21 (Linear Fourier score)**.** *For $f : \mathbb{F}_q^n \to \mathbb{F}_q$ and $\alpha \in \mathbb{F}_q^n$, define the linear Fourier score of f at α by*

$$S_\alpha(f) := \sum_{c\in\mathbb{F}_q^\times} \widehat{\varphi_c}(c\alpha),$$

where $\{\varphi_c\}_{c\in\mathbb{F}_q^\times}$ is the Fourier expansion of f.

Lemma 1.1.22 (Distance to linearity in Fourier form)**.** *Let $f : \mathbb{F}_q^n \to \mathbb{F}_q$ have Fourier expansion $\{\varphi_c\}_{c\in\mathbb{F}_q^\times}$. Then*

$$\delta(f, \mathcal{LIN}) = 1 - \frac{1}{q}\left(1 + \max_{\alpha\in\mathbb{F}_q^n} S_\alpha(f)\right).$$

Moreover, for every $\alpha \in \mathbb{F}_q^n$, the score $S_\alpha(f)$ is real and satisfies $-1 \le S_\alpha(f) \le q-1$.

Proof. Apply Theorem 1.1.19 with $g = \ell_\alpha$ and use Theorem 1.1.20. This gives

$$\delta(f, \ell_\alpha) = 1 - \frac{1}{q}\left(1 + S_\alpha(f)\right).$$

Taking the minimum over α gives the claimed formula for $\delta(f, \mathcal{LIN})$. For fixed α, the same formula expresses $S_\alpha(f)$ as $q(1 - \delta(f, \ell_\alpha)) - 1$. Hence $S_\alpha(f)$ is real, and since $0 \le \delta(f, \ell_\alpha) \le 1$, we get $-1 \le S_\alpha(f) \le q - 1$. □

Definition 1.1.23 (Finite-field BLR test)**.** *Given oracle access to $f : \mathbb{F}_q^n \to \mathbb{F}_q$, the finite-field BLR verifier $\mathsf{V}_{\mathsf{BLR}}^f$ samples $a, b \in \mathbb{F}_q^\times$ and $x, y \in \mathbb{F}_q^n$ independently and uniformly at random, and accepts iff*

$$af(x) + bf(y) = f(ax + by).$$

That is,

$$\mathsf{V}_{\mathsf{BLR}}^f = 1 \quad\Longleftrightarrow\quad af(x) + bf(y) = f(ax + by).$$

Lemma 1.1.24 (Completeness)**.** *If $f \in \mathcal{LIN}$, then*

$$\Pr[\mathsf{V}_{\mathsf{BLR}}^f = 1] = 1.$$

Proof. If $f = \ell_\alpha$ for some $\alpha \in \mathbb{F}_q^n$, then for every $a, b \in \mathbb{F}_q^\times$ and $x, y \in \mathbb{F}_q^n$,

$$f(ax + by) = \langle\alpha, ax + by\rangle = a\langle\alpha, x\rangle + b\langle\alpha, y\rangle = af(x) + bf(y).$$

Thus the verifier accepts for every choice of randomness. □

Lemma 1.1.25 (Exact BLR acceptance formula)**.** *Let $f : \mathbb{F}_q^n \to \mathbb{F}_q$. Then*

$$\Pr[\mathsf{V}_{\mathsf{BLR}}^f = 1] = \frac{1}{q}\left(1 + \frac{1}{(q-1)^2}\sum_{\alpha\in\mathbb{F}_q^n} S_\alpha(f)^3\right).$$

Proof. By Theorem 1.1.14, for fixed a, b, x, y,

$$\mathbb{1}[af(x) + bf(y) = f(ax + by)] = \frac{1}{q} \sum_{c \in \mathbb{F}_q} \omega_p^{\mathrm{Tr}(c(af(x)+bf(y)-f(ax+by)))}.$$

Taking expectation gives

$$\Pr[\mathsf{V}_{\mathsf{BLR}}^f = 1] = \frac{1}{q} + \frac{1}{q} \mathbb{E}_{a,b,x,y} \left[\sum_{c \in \mathbb{F}_q^\times} \varphi_{ca}(x) \varphi_{cb}(y) \varphi_{-c}(ax + by) \right].$$

Expand the three complex-valued functions into their Fourier expansions:

$$\begin{aligned}
&\mathbb{E}_{a,b,x,y} \left[\sum_{c \in \mathbb{F}_q^\times} \varphi_{ca}(x) \varphi_{cb}(y) \varphi_{-c}(ax + by) \right] \\
&= \mathbb{E}_{a,b} \left[\sum_{c \in \mathbb{F}_q^\times} \sum_{\alpha, \beta, \gamma \in \mathbb{F}_q^n} \widehat{\varphi_{ca}}(\alpha) \widehat{\varphi_{cb}}(\beta) \widehat{\varphi_{-c}}(\gamma) \mathbb{E}_{x,y} \left[\chi_\alpha(x) \chi_\beta(y) \chi_\gamma(ax + by) \right] \right].
\end{aligned}$$

Since

$$\chi_\gamma(ax + by) = \chi_{a\gamma}(x) \chi_{b\gamma}(y),$$

Theorem 1.1.8 implies that the inner expectation is 1 exactly when

$$\alpha + a\gamma = 0 \quad \text{and} \quad \beta + b\gamma = 0,$$

and is 0 otherwise. Therefore the last display equals

$$\mathbb{E}_{a,b} \left[\sum_{c \in \mathbb{F}_q^\times} \sum_{\alpha \in \mathbb{F}_q^n} \widehat{\varphi_{ca}}(\alpha) \widehat{\varphi_{cb}}(a^{-1} b\alpha) \widehat{\varphi_{-c}}(-a^{-1}\alpha) \right].$$

Substitute $\alpha = ca\eta$. Then this becomes

$$\mathbb{E}_{a,b} \left[\sum_{c \in \mathbb{F}_q^\times} \sum_{\eta \in \mathbb{F}_q^n} \widehat{\varphi_{ca}}(ca\eta) \widehat{\varphi_{cb}}(cb\eta) \widehat{\varphi_{-c}}(-c\eta) \right].$$

As a, b, c range over $\mathbb{F}_q^\times$, so do ca, cb, and $-c$. Hence this equals

$$\frac{1}{(q-1)^2} \sum_{\eta \in \mathbb{F}_q^n} \left(\sum_{d \in \mathbb{F}_q^\times} \widehat{\varphi_d}(d\eta) \right)^3 = \frac{1}{(q-1)^2} \sum_{\eta \in \mathbb{F}_q^n} S_\eta(f)^3.$$

Substituting into the expression for $\Pr[\mathsf{V}_{\mathsf{BLR}}^f = 1]$ proves the claim. □

Lemma 1.1.26 (Second moment bound)**.** *For every $f : \mathbb{F}_q^n \to \mathbb{F}_q$, one has*

$$\sum_{\alpha \in \mathbb{F}_q^n} S_\alpha(f)^2 \leq (q-1)^2.$$

Proof. Expanding the square gives

$$\sum_{\alpha\in\mathbb{F}_q^n} S_\alpha(f)^2 = \sum_{a,b\in\mathbb{F}_q^\times} \sum_{\alpha\in\mathbb{F}_q^n} \widehat{\varphi_a}(a\alpha)\widehat{\varphi_b}(b\alpha).$$

For fixed $a, b \in \mathbb{F}_q^\times$, Cauchy-Schwarz and Theorem 1.1.11 give

$$\begin{aligned}\left|\sum_{\alpha\in\mathbb{F}_q^n} \widehat{\varphi_a}(a\alpha)\widehat{\varphi_b}(b\alpha)\right| &\le \left(\sum_{\alpha\in\mathbb{F}_q^n} |\widehat{\varphi_a}(a\alpha)|^2\right)^{1/2} \left(\sum_{\alpha\in\mathbb{F}_q^n} |\widehat{\varphi_b}(b\alpha)|^2\right)^{1/2} \\ &= 1.\end{aligned}$$

The equality uses that multiplication by a and by b permutes $\mathbb{F}_q^n$, and that Parseval plus $|\varphi_a(x)| = |\varphi_b(x)| = 1$ for every x make both squared L_2 norms equal to 1. Summing over the $(q-1)^2$ choices of (a, b) gives the result. $\square$

Theorem 1.1.27 (Soundness of the finite-field BLR test)**.** *For every function $f : \mathbb{F}_q^n \to \mathbb{F}_q$, one has*

$$\Pr[\mathsf{V}_{\mathsf{BLR}}^f = 1] \le 1 - \delta(f, \mathcal{LIN}).$$

Proof. Let $S_\alpha(f)$ be as in Theorem 1.1.21, and let

$$M := \max_{\alpha\in\mathbb{F}_q^n} S_\alpha(f).$$

By Theorem 1.1.22,

$$1 - \delta(f, \mathcal{LIN}) = \frac{1}{q}(1 + M).$$

By Theorem 1.1.22, each $S_\alpha(f)$ is real and $S_\alpha(f) \le M$. Therefore $S_\alpha(f)^3 \le M S_\alpha(f)^2$ for every α, because $S_\alpha(f)^2 \ge 0$. Using Theorem 1.1.25 and Theorem 1.1.26,

$$\begin{aligned}\Pr[\mathsf{V}_{\mathsf{BLR}}^f = 1] &= \frac{1}{q}\left(1 + \frac{1}{(q-1)^2}\sum_{\alpha\in\mathbb{F}_q^n} S_\alpha(f)^3\right) \\ &\le \frac{1}{q}\left(1 + \frac{M}{(q-1)^2}\sum_{\alpha\in\mathbb{F}_q^n} S_\alpha(f)^2\right) \\ &\le \frac{1}{q}(1 + M) \\ &= 1 - \delta(f, \mathcal{LIN}).\end{aligned}$$

$\square$

Alternative proof via Gowers–Hatami. First denote

$$X := \mathbb{F}_q^n, \qquad A := \mathbb{F}_q^\times, \qquad \omega := e^{2\pi i/p}.$$

Define a group G on the set $X \times A$ with the binary operation

$$(x, a) \star (y, b) := (b^{-1}x + a^{-1}y,\ ab).$$

Let F_f be the unitary matrix valued lift $F_f : G \to U(\mathbb{C}^{q-1})$ given by the diagonal matrix

$$F_f(x, a) = D(af(x)) := \sum_{c \in A} \omega^{\operatorname{Tr} caf(x)} |c\rangle\langle c|,$$

where $\{|c\rangle\}$ denotes the orthonormal basis vectors of a Hilbert space $\mathcal{H} \cong \mathbb{C}^{q-1}$, the inner product between operators is $\langle A, B\rangle_\sigma = \operatorname{Tr}(A^* B\sigma)$, and $\sigma = \mathbb{I}/(q-1)$. By Lemma 2.4.7, since f is rejected with probability ε, the lift F_f is a $(\frac{q}{q-1}\varepsilon, \frac{\mathbb{I}}{q-1})$ approximate representation of $(G, \star)$.

Then by Lemma 2.4.9, there exists a linear function $\ell : \mathbb{F}_q^n \to \mathbb{F}_q$ whose relative Hamming distance to f is bounded by

$$\delta(f, \ell) \le \frac{q-1}{q} \cdot \frac{q}{q-1}\varepsilon = \varepsilon.$$

This proves the claim

$$\delta(f, \mathcal{LIN}) := \min_\alpha \delta(f, \ell_\alpha) \le \delta(f, \ell) \le \varepsilon.$$

Miraculously, the loose factor $\frac{q}{q-1}$ in the approximate-representation parameter cancels the tighter factor $\frac{q-1}{q}$ of Lemma 2.4.9.

In Lean this route is formalized as `gh_blr_soundness_acceptance`, which proves `acceptanceProbabilityBLR f ≤ 1 - distanceToLinear f` (the statement of `blr_soundness`), valid for $\frac{q}{q-1}\varepsilon \le 1$; the unconditional statement is the direct proof above, since for $\frac{q}{q-1}\varepsilon > 1$ the third-moment extraction's $\max \ge 0$ step fails. □

Theorem 1.1.28 (Counterexample to the proposed cubic lower bound)**.** *The inequality*

$$(1 - \delta(f, \mathcal{LIN}))^3 \le \Pr[\mathsf{V}^f_{\mathsf{BLR}} = 1]$$

does not hold for all functions $f : \mathbb{F}_q^n \to \mathbb{F}_q$ with the finite-field BLR verifier from Theorem 1.1.23.

Proof. It suffices to give a concrete counterexample. Take $q = 2$ and $n = 1$, so the domain and codomain are both $\mathbb{F}_2$. Let

$$f : \mathbb{F}_2 \to \mathbb{F}_2, \qquad f(0) = 1, \quad f(1) = 1.$$

The homogeneous linear maps $\mathbb{F}_2 \to \mathbb{F}_2$ are exactly the zero map and the identity map. The function f agrees with the zero map at no point, and it agrees with the identity map only at the point 1. Hence

$$\delta(f, \mathcal{LIN}) = \frac{1}{2}, \qquad 1 - \delta(f, \mathcal{LIN}) = \frac{1}{2}.$$

For the verifier, the only possible nonzero field elements are $a = b = 1$. Thus the verifier accepts on a pair (x, y) exactly when

$$f(x) + f(y) = f(x + y).$$

Since f is constantly 1, the left-hand side is $1 + 1 = 0$ in $\mathbb{F}_2$, while the right-hand side is 1. Therefore the verifier rejects for every choice of x, y, and

$$\Pr[\mathsf{V}^f_{\mathsf{BLR}} = 1] = 0.$$

But

$$(1 - \delta(f, \mathcal{LIN}))^3 = \left(\frac{1}{2}\right)^3 = \frac{1}{8} > 0 = \Pr[\mathsf{V}^f_{\mathsf{BLR}} = 1].$$

This contradicts the proposed lower bound. □

Theorem 1.1.29 (Lower bound from agreement with a linear function)**.** *Let* $f : \mathbb{F}_q^n \to \mathbb{F}_q$*. Suppose there exists* $\ell \in \mathcal{LIN}$ *such that*

$$\Pr_{x \in \mathbb{F}_q^n}[f(x) = \ell(x)] \geq 1 - \varepsilon.$$

Then

$$\Pr[\mathsf{V}_{\mathsf{BLR}}^f = 1] \geq 1 - 3\varepsilon + 2\varepsilon^2.$$

In particular,

$$\Pr[\mathsf{V}_{\mathsf{BLR}}^f = 1] \geq 1 - 3\delta(f, \mathcal{LIN}) + 2\delta(f, \mathcal{LIN})^2.$$

Proof. Let

$$G := \{z \in \mathbb{F}_q^n : f(z) = \ell(z)\}$$

be the agreement set, and let $B := \mathbb{F}_q^n \setminus G$ be the bad set. Write

$$\eta := \Pr_z[z \in B].$$

By assumption, $\eta \leq \varepsilon$.

The verifier samples $a, b \in \mathbb{F}_q^\times$ and $x, y \in \mathbb{F}_q^n$, and queries

$$x, \qquad y, \qquad ax + by.$$

For fixed $a, b \in \mathbb{F}_q^\times$, the random variables x, y, and $ax + by$ are each uniform on $\mathbb{F}_q^n$. Indeed, x and y are uniform by definition, and $ax + by$ is uniform because, after fixing a, b, and x, the map

$$y \mapsto ax + by$$

is a bijection of $\mathbb{F}_q^n$.

Therefore

$$\Pr[x \in B] = \Pr[y \in B] = \Pr[ax + by \in B] = \eta.$$

Moreover the three queried points are pairwise independent. For fixed $a, b \in \mathbb{F}_q^\times$, each of the maps

$$(x, y) \mapsto (x, y), \qquad (x, y) \mapsto (x, ax + by), \qquad (x, y) \mapsto (y, ax + by)$$

is a bijection from $\mathbb{F}_q^n \times \mathbb{F}_q^n$ to itself. Hence

$$\Pr[x \in B,\ y \in B] = \Pr[x \in B,\ ax + by \in B] = \Pr[y \in B,\ ax + by \in B] = \eta^2.$$

By inclusion–exclusion and the trivial bound

$$\Pr[x \in B,\ y \in B,\ ax + by \in B] \leq \Pr[x \in B,\ y \in B] = \eta^2,$$

we get

$$\Pr[x \in B \text{ or } y \in B \text{ or } ax + by \in B] \leq 3\eta - 3\eta^2 + \eta^2 = 3\eta - 2\eta^2.$$

On the complementary event, the verifier sees the same three values as it would see from ℓ:

$$f(x) = \ell(x), \qquad f(y) = \ell(y), \qquad f(ax + by) = \ell(ax + by).$$

Since ℓ is linear,

$$\ell(ax + by) = a\ell(x) + b\ell(y).$$

Thus

$$f(ax+by) = af(x) + bf(y),$$

so the verifier accepts. Hence the acceptance probability is at least the probability of the complementary event:

$$\Pr[\mathsf{V}^f_{\mathsf{BLR}} = 1] \geq 1 - 3\eta + 2\eta^2.$$

If $\varepsilon \geq 1/2$, then $1 - 3\varepsilon + 2\varepsilon^2 \leq 0$, so the desired bound is trivial. Otherwise $\varepsilon \leq 1/2$, and since $\eta \leq \varepsilon$, the function $1 - 3t + 2t^2$ is decreasing on the interval containing both η and ε. This gives

$$\Pr[\mathsf{V}^f_{\mathsf{BLR}} = 1] \geq 1 - 3\varepsilon + 2\varepsilon^2.$$

Taking ℓ to be a closest linear function gives the final statement with $\varepsilon = \delta(f, \mathcal{LIN})$. □

Theorem 1.1.30 (Boolean tightness for subspace error sets)**.** *The bound in Theorem 1.1.29 is tight for the Boolean BLR test for infinitely many values of ε. More precisely, let $q = 2$ and let $B \leq \mathbb{F}_2^n$ be a linear subspace of density $\varepsilon = 2^{-k}$. Define*

$$f : \mathbb{F}_2^n \to \mathbb{F}_2, \qquad f(x) = \begin{cases} 1 & \text{if } x \in B, \\ 0 & \text{if } x \notin B. \end{cases}$$

Then $\delta(f, \mathcal{LIN}) = \varepsilon$ and

$$\Pr[\mathsf{V}^f_{\mathsf{BLR}} = 1] = 1 - 3\varepsilon + 2\varepsilon^2.$$

Proof. The zero linear function disagrees with f exactly on B, so $\delta(f, \mathcal{LIN}) \leq \varepsilon$. If $m : \mathbb{F}_2^n \to \mathbb{F}_2$ is any nonzero linear function, then m is balanced. Thus m agrees with the indicator of B on at most half of the domain, whereas the zero linear function agrees on a $1-\varepsilon$ fraction of the domain. For $\varepsilon \leq 1/2$, this shows that the closest linear function is the zero function, so $\delta(f, \mathcal{LIN}) = \varepsilon$.

Since $q = 2$, the verifier has $a = b = 1$ and accepts iff

$$f(x) + f(y) = f(x+y).$$

Because B is a subspace, among the three points x, y, and $x+y$, the number lying in B is never exactly two. Therefore the verifier rejects exactly when at least one of the three queried points lies in B.

The three events

$$x \in B, \qquad y \in B, \qquad x+y \in B$$

each have probability ε, and each pair has probability ε^2. Since B is a subspace, the triple event also has probability ε^2: if $x, y \in B$, then $x + y \in B$. Hence

$$\Pr[\mathsf{V}^f_{\mathsf{BLR}} = 0] = 3\varepsilon - 3\varepsilon^2 + \varepsilon^2 = 3\varepsilon - 2\varepsilon^2.$$

Taking complements gives

$$\Pr[\mathsf{V}^f_{\mathsf{BLR}} = 1] = 1 - 3\varepsilon + 2\varepsilon^2.$$

□

Chapter 2

Gowers Hatami theorem

2.1 Preliminaries

Definition 2.1.1 (σ inner product.). *The σ inner product between matrices A and B is defined as*

$$\langle A, B\rangle_\sigma = \mathrm{Tr}(A^* B\sigma), \tag{2.1}$$

where $\sigma \geq 0$ and A^* denotes conjugate-transpose.

Definition 2.1.2 ((ϵ, σ)-representation.). *Given a finite group G, an integer $d \geq 1, \varepsilon \geq 0$, and a d-dimensional positive semidefinite matrix σ with trace 1 , an (ε, σ)-representation of G is a map $f : G \to U_d(\mathbb{C})$, to the $d \times d$ unitary matrices, such that*

$$\mathbb{E}_{x,y\in G}\Re\left(\left\langle f(x)^* f(y), f\left(x^{-1}y\right)\right\rangle_\sigma\right) \geq 1 - \varepsilon, \tag{2.2}$$

where $\Re$ denotes taking real part of the inner product.

Proposition 2.1.3 (σ-norm Square). *Let σ be positive semidefinite and let A, B be matrices of the same dimension. Then*

$$\|A - B\|_\sigma^2 = \|A\|_\sigma^2 + \|B\|_\sigma^2 - 2\,\Re\langle A, B\rangle_\sigma.$$

Proof. Expanding by linearity of the inner product,

$$\|A - B\|_\sigma^2 = \Re\langle A - B, A - B\rangle_\sigma = \|A\|_\sigma^2 + \|B\|_\sigma^2 - \Re\langle A, B\rangle_\sigma - \Re\langle B, A\rangle_\sigma.$$

Since σ is Hermitian, $\langle B, A\rangle_\sigma = \overline{\langle A, B\rangle_\sigma}$, so the two cross terms have the same real part, yielding $-2\,\Re\langle A, B\rangle_\sigma$. □

2.2 Representation theory

Definition 2.2.1 (Unitary Representation). *A* unitary representation *of G is a representation to unitary matrices $\rho : G \to U(V)$.*

Definition 2.2.2 (Regular representation). *Let $R : G \to \mathbb{C}[G]$ be a regular representation where $\{|g\rangle, g \in G\}$ forms a basis of $\mathbb{C}[G]$. The representation is given by*

$$R(x)|g\rangle = |xg\rangle. \tag{2.3}$$

It is possible to decompose any representation of a finite group into direct sums over irreps.

Theorem 2.2.3 (Maschke's theorem). *Let G be a finite group and let $\rho : G \to \mathrm{GL}(V)$ be a finite-dimensional representation over $\mathbb{C}$. Then ρ is completely reducible. In particular, there exists a decomposition*

$$\rho \cong \bigoplus_m r_m\, \rho_m, \tag{2.4}$$

where $\{\rho_m\}$ is a set of inequivalent irreducible representations of G, and $r_m \in \mathbb{Z}^+$ are the multiplicities.

Proof sketch. Maschke's theorem is partially formalized in `Mathlib.RepresentationTheory.Maschke` but incomplete. Jonathan's team is formalizing this. □

Proposition 2.2.4 (Decomposition of the regular representation). *Let $R : G \to \mathrm{GL}(\mathbb{C}[G])$ be the regular representation of a finite group G. Then*

$$R \cong \bigoplus_{\rho \in \widehat{G}} d_\rho\, \rho, \tag{2.5}$$

where $\widehat{G}$ is a complete set of inequivalent irreducible representations of G, and $d_\rho = \dim(V_\rho)$ is the dimension of ρ.

Proof sketch. This can be proven using Maschke's theorem in 2.2.3 to decompose into irreps with multiplicity. Then use character orthogonality theorem in mathlib `FDRep.char_orthonormal` to show the multiplicity equals irrep dimension. A more formal statement is given by the Peter-Weyl theorem. □

Proposition 2.2.5 (Character of regular representation.). *Let G be a finite group and $\sum_\rho$ denotes summing over the complete set of inequivalent irreps $\hat{G}$*

$$\sum_{\rho \in \hat{G}} d_\rho \operatorname{Tr}(\rho(x)) = |G|\delta_{x,e}. \tag{2.6}$$

Proof sketch. The proof uses decomposition of regular representation and definition of character. □

Definition 2.2.6 (Group fourier transform). *Let $f : G \to U_d(\mathbb{C})$ be a map and $\rho : G \to U_{d_\rho}(\mathbb{C})$ be a unitary irrep. The fourier transform of f at the representation ρ is denoted by $\hat{f}$:*

$$\hat{f}(\rho) = \mathbb{E}_{x \in G} f(x) \otimes \overline{\rho(x)}, \tag{2.7}$$

where $\overline{\rho(x)}$ denotes complex conjugate of $\rho(x)$.

2.3 Gowers Hatami Theorem

Theorem 2.3.1 (Gowers-Hatami). *Let G be a finite group, $\varepsilon \geq 0$, and $f : G \to U_d(\mathbb{C})$ an (ε, σ)-representation of G. Then there exists a $d' \geq d$, an isometry $V : \mathbb{C}^d \to \mathbb{C}^{d'}$, and a representation $g : G \to U_{d'}(\mathbb{C})$ such that*

$$\mathbb{E}_{x \in G} \|f(x) - V^* g(x) V\|_\sigma^2 \leq 2\varepsilon. \tag{2.8}$$

Proof sketch. The theorem is due to Gowers and Hatami [**?**]; the constructive proof below follows Vidick's lecture notes, Theorem 2.3 [**?**]. In particular, the proof is constructive. The isometry is defined as $V : \mathbb{C}^d \to \mathbb{C}^d \otimes \left(\oplus_\rho \mathbb{C}^{d_\rho} \otimes \mathbb{C}^{d_\rho}\right)$ by

$$Vu = \bigoplus_\rho d_\rho^{1/2} \sum_{i=1}^{d_\rho} \left(\hat{f}(\rho)\,(u \otimes e_i)\right) \otimes e_i. \tag{2.9}$$

And the exact representation is defined as

$$g(x) = \bigoplus_\rho \left(I_d \otimes I_{d_\rho} \otimes \rho(x)\right). \tag{2.10}$$

The proof requires the following two ingredients:

1. V is an isometry:
$$V^*V = \mathbb{I}_d \tag{2.11}$$

2. The isometry "averages" over group multiplication. For any $x \in G$,
$$V^*g(x)V = \mathbb{E}_z f(z)^* f(zx). \tag{2.12}$$

Both ingredients follow from the key identity over the regular representation in Eq. (2.6). □

Next, we will give a different proof that does not directly decompose into irreps but instead uses the action of the regular representation.

Lemma 2.3.2 (Gowers-Hatami Pair-Correlation)**.** *Let G be a finite group, $\varepsilon \geq 0$, and $\rho : G \to U(\mathcal{H})$ an (ε, σ)-representation of G on space $\mathcal{H}$. Define the pair-correlation of ρ as:*

$$\mathcal{C}(\rho) := \mathbb{E}_{x,y\in G} \Re\langle \rho(y)\rho(x), \rho(yx)\rangle_\sigma. \tag{2.13}$$

Then there exists an isometry $V : \mathcal{H} \to \mathcal{H}'$ to a different space $\mathcal{H}'$, and an exact representation $\rho' : G \to U(\mathcal{H}')$ such that the average correlation between ρ and ρ' is exactly the pair-correlation, which is bounded by $1 - \varepsilon$:

$$\mathbb{E}_{x\in G} \Re\langle \rho(x), V^*\rho'(x)V\rangle_\sigma = \mathcal{C}(\rho) \geq 1 - \varepsilon. \tag{2.14}$$

Proof. Here $\mathcal{H}' = L(G, \mathcal{H})$ is the space of functions $f : G \to \mathcal{H}$.

Let the isometry be defined by the action of the approximate representation on a state $u \in \mathcal{H}$ as

$$V : \quad \mathcal{H} \to L(G, \mathcal{H}), \tag{2.15}$$
$$V(u) : \quad G \to \mathcal{H}, \quad x \mapsto \rho(x)(u), \quad \forall x \in G. \tag{2.16}$$

The adjoint map V^* is defined as the unique map such that for all $u \in \mathcal{H}$ and $f \in L(G, \mathcal{H})$,

$$\langle V(u), f\rangle_{L(G,\mathcal{H})} = \langle u, V^*(f)\rangle_{\mathcal{H}}. \tag{2.17}$$

In particular, V^* is explicitly given by the group average:

$$V^* : \quad L(G, \mathcal{H}) \to \mathcal{H}, \tag{2.18}$$
$$V^*(f) \quad = \mathbb{E}_{x\in G}\left[\rho^*(x) f(x)\right]. \tag{2.19}$$

The exact representation is given by the right regular representation on $L(G, \mathcal{H})$:

$$\begin{aligned} \rho' : \quad & G \to U(L(G,\mathcal{H})), && (2.20)\\ (\rho'(x)f)(y) \quad & = f(yx), \quad \forall x, y \in G. && (2.21) \end{aligned}$$

V is an isometry because $\rho(x)$ is unitary, yielding:

$$V^*V(u) = \mathbb{E}_{x\in G}\left[\rho^*(x)\rho(x)(u)\right] = \mathbb{E}_{x\in G}[u] = u. \tag{2.22}$$

The compression of the exact representation "averages" over group multiplication. For any $u \in \mathcal{H}$:

$$\begin{aligned} V^*\rho'(x)V(u) \quad &= \quad \mathbb{E}_{y\in G}\left[\rho^*(y)(\rho'(x)\rho(y))(u)\right] \\ &= \quad \mathbb{E}_{y\in G}\left[\rho^*(y)\rho(yx)(u)\right]. \end{aligned} \tag{2.23}$$

Substituting the average property from Eq. 2.23 into the inner product, and using the unitarity of $\rho(y)$ to rewrite the adjoints ($\langle A, B^*C\rangle_\sigma = \langle BA, C\rangle_\sigma$), we obtain:

$$\begin{aligned} \mathbb{E}_{x\in G}\Re\langle\rho(x), V^*\rho'(x)V\rangle_\sigma \quad &= \quad \mathbb{E}_{x,y\in G}\Re\langle\rho(x), \rho^*(y)\rho(yx)\rangle_\sigma \\ &= \quad \mathbb{E}_{x,y\in G}\Re\langle\rho(y)\rho(x), \rho(yx)\rangle_\sigma \\ &= \quad \mathcal{C}(\rho). \end{aligned} \tag{2.24}$$

By the definition of an (ε, σ)-approximate representation, the pair-correlation inherently satisfies $\mathcal{C}(\rho) \geq 1 - \varepsilon$, which concludes the proof. $\square$

Theorem 2.3.3 (Gowers-Hatami-2)**.** *Let G be a finite group, $\varepsilon \geq 0$, and $\rho : G \to U(\mathcal{H})$ an (ε, σ)-representation of G on space $\mathcal{H} = \mathbb{C}^d$. Then there exists an isometry $V : \mathcal{H} \to \mathcal{H}'$ to a different space $\mathcal{H}'$, and a representation $\rho' : G \to \mathcal{H}'$ such that*

$$\mathbb{E}_{x\in G}\left\|\rho(x) - V^*\rho'(x)V\right\|_\sigma^2 \leq 2\varepsilon. \tag{2.25}$$

Proof. A regular-representation proof of the Gowers-Hatami theorem appears in Morel's slides [**?**], following the original Gowers-Hatami theorem [**?**].

The rest of the proof follows by expanding the σ-norm, bounding the norm of $\mathbb{E}_{x\in G}V^*\rho'(x)V$ and applying the definition of (ε, σ)-representation. In particular, we have

$$\begin{aligned} & \mathbb{E}_{x\in G}\left\|\rho(x) - V^*\rho'(x)V\right\|_\sigma^2 && (2.26)\\ = \quad & \mathbb{E}_{x\in G}\|\rho(x)\|_\sigma^2 + \mathbb{E}_{x\in G}\|V^*\rho'(x)V\|_\sigma^2 - 2\mathbb{E}_{x\in G}\Re\langle\rho(x), V^*\rho'(x)V\rangle_\sigma \end{aligned}$$

Because ρ is unitary and $\mathrm{Tr}\sigma = 1$, $\mathbb{E}_{x\in G}\|\rho(x)\|_\sigma^2 = 1$. For the second term, recall from Eq. (2.23) that the compression is the group average $V^*\rho'(x)V = \mathbb{E}_{y\in G}\rho^*(y)\rho(yx)$. Each summand $\rho^*(y)\rho(yx)$ is a product of unitaries, hence unitary, so $\|\rho^*(y)\rho(yx)\|_\sigma = 1$. By convexity of the σ-norm—the triangle inequality applied to the average—we obtain

$$\|V^*\rho'(x)V\|_\sigma = \left\|\mathbb{E}_{y\in G}\rho^*(y)\rho(yx)\right\|_\sigma \leq \mathbb{E}_{y\in G}\|\rho^*(y)\rho(yx)\|_\sigma = 1, \tag{2.27}$$

hence $\mathbb{E}_{x\in G}\|V^*\rho'(x)V\|_\sigma^2 \leq 1$.

Therefore using the bounds in Eq. 2.27 and Eq. 2.24 by applying Lemma 2.3.2, we have

$$\mathbb{E}_{x\in G}\left\|\rho(x) - V^*\rho'(x)V\right\|_\sigma^2 \leq 2 - 2(1-\varepsilon) = 2\varepsilon. \tag{2.28}$$

$\square$

2.4 Gowers-Hatami for BLR linearity test over $\mathbb{F}_q$

Assume $q = p^s$ be a prime with power $s \in \mathbb{Z}^+$, $n \geq 1$, and we set

$$X := \mathbb{F}_q^n, \qquad A := \mathbb{F}_q^\times, \qquad \omega := e^{2\pi i/p}.$$

Definition 2.4.1 (BLR verifier and distance)**.** *Given $f : X \to \mathbb{F}_q$, the verifier $\mathsf{V}_{\mathrm{BLR}}^f$ samples $a, b \in A$ and $x, y \in X$ uniformly and accepts iff $af(x) + bf(y) = f(ax + by)$. Let the test rejection probability be bounded by*

$$\Pr_{a,b,x,y}\big[af(x) + bf(y) \neq f(ax + by)\big] \leq \varepsilon.$$

For $\alpha \in X$, write $\ell_\alpha(x) := \langle \alpha, x \rangle$, $\delta_\alpha := \Pr_x[f(x) \neq \ell_\alpha(x)]$, and $\delta(f, \mathcal{LIN}) := \min_\alpha \delta_\alpha$.

At the end of the section, we prove the BLR soundness using Gowers-Hatami theorem. First notice that linearity implies two properties:

1. Additivity $f(x + y) = f(x) + f(y), \quad \forall x, y \in X$.

   Note that for q prime additivity is actually equivalent to linearity. However this is not the case for prime powers $q = p^s$ for $s > 1$.

2. Homogeneity $f(ax) = af(x), \quad \forall x \in X,\ a \in A$.

   For $q = 2$, the two properties are redundant. For $q > 2$, turns out testing homogeneity is crucial for a test to have a high soundness.

2.4.1 The twisted group G

Define a group G by the set $X \times A$ with the binary operation

$$(x, a) \star (y, b) := (b^{-1}x + a^{-1}y,\ ab). \tag{2.29}$$

Lemma 2.4.2 (G is an abelian group)**.** *The operation $\star$ in (2.29) makes G into an abelian group with identity $e = (0, 1)$ and inverses $(x, a)^{-1} = (-a^2x, a^{-1})$.*

Proof. Associativity. For $(x, a), (y, b), (z, c) \in G$,

$$((x, a) \star (y, b)) \star (z, c) = (b^{-1}x + a^{-1}y,\ ab) \star (z, c) = (b^{-1}c^{-1}x + a^{-1}c^{-1}y + a^{-1}b^{-1}z,\ abc),$$

and an identical computation gives the same expression for $(x, a) \star ((y, b) \star (z, c))$.

Commutativity. $(y, b) \star (x, a) = (a^{-1}y + b^{-1}x,\ ba) = (b^{-1}x + a^{-1}y,\ ab) = (x, a) \star (y, b)$.

Identity. For any $(y, b) \in G$, $(0, 1) \star (y, b) = (b^{-1} \cdot 0 + 1^{-1} \cdot y,\ 1 \cdot b) = (y, b)$.

Inverse. For any $(x, a) \in G$,

$$(x, a) \star (-a^2x, a^{-1}) = ((a^{-1})^{-1} \cdot x + a^{-1} \cdot (-a^2x),\ a \cdot a^{-1}) = (ax - ax,\ 1) = (0, 1).$$

□

It is at first surprising to come up with $\star$ operation, but G is isomorphic to the direct product group.

Proposition 2.4.3 (Group isomorphism to product group)**.** *Let $(G, \star)$ be an abelian group defined by Equation 2.29. The map*

$$\Psi : G \to (X, +) \times (A, \cdot), \qquad \Psi(x, a) := (ax, a), \tag{2.30}$$

is a group isomorphism, where the codomain is the direct product of $(X, +)$ and $(A, \cdot)$ with operation $\star_\Psi : (x,\ a) \star_\Psi (y,\ b) = (x + y,\ ab)$.

Proof. Compute

$$\Psi\big((x, a) \star (y, b)\big) = \Psi(b^{-1}x + a^{-1}y,\ ab) = \big(ab \cdot (b^{-1}x + a^{-1}y),\ ab\big) = (ax + by,\ ab),$$

which equals $\Psi(x, a) \star_\Psi \Psi(y, b) = (ax + by,\ ab)$. The map Ψ is bijective with inverse $\Psi^{-1}(y, b) = (b^{-1}y, b)$. □

Remark (formalization). The isomorphism Ψ is used here only to motivate and compute the characters. The Lean development does *not* formalize Ψ as a group isomorphism: it equips G with its group structure (Lemma 2.4.2) and defines the characters $\chi_{\alpha,k}$ (Proposition 2.4.5) directly on the twisted group, proving the homomorphism property by a direct computation rather than by pulling back through Ψ. Hence Proposition 2.4.3 carries no `\lean` link.

The isomorphism gives us a easy way to compute its irreducible representations which forms the Pontryagin dual group (the set of irrep characters for an abelian group).

Fact 2.4.4 (Pontryagin dual over direct product)**.** *Let G_1, G_2 be finite abelian groups. Let $H = G_1 \times G_2$ be their direct product. Then the Pontryagin dual formed by all the irreps. is $\hat{H} = \hat{G_1} \times \hat{G_2}$.*

Proposition 2.4.5 (Group characters)**.** *The group $(G, \star)$ has the Pontryagin dual character group*

$$\hat{G} = \{\chi_{\alpha,k}\}_{\alpha \in X, k \in [0,\cdots,q-2]}, \quad \chi_{\alpha,k}(x, a) = \omega^{\mathrm{Tr}\langle \alpha, ax \rangle} \psi_k(a), \tag{2.31}$$

where $\psi_k(a) = e^{\frac{i2\pi mk}{q-1}}$ and $a = g_0^m$ for a fixed multiplicative generator $g_0 \in A$.

Proof. By Proposition 2.4.3, $(G, \star) \cong (X, +) \times (A, \cdot)$ via $\Psi(x, a) = (ax, a)$. First we consider the characters of the product group. Using Fact. 2.4.4, we have

$$\widehat{(X, +) \times (A, \cdot)} = \hat{X} \times \hat{A}.$$

The character groups of X and A are

$$\hat{X} = \{\phi_\alpha(y) = \omega^{\mathrm{Tr}\langle \alpha, y \rangle}\}_{\alpha \in X}, \tag{2.32}$$

$$\hat{A} = \{\psi_k(b) = e^{\frac{i2\pi mk}{q-1}}\}_{k \in [0,\cdots,q-2]}, \tag{2.33}$$

where $b = g_0^m$ for a fixed multiplicative generator $g_0 \in A$. Here the field trace $\mathrm{Tr} : \mathbb{F}_q \to \mathbb{F}_p$ is defined by $\mathrm{Tr}(x) = x + x^p + x^{p^2} + \cdots + x^{p^{s-1}}$ for $x \in \mathbb{F}_q$ (see Definition 1.1 in Chapter 1).Therefore, $\widehat{(X, +) \times (A, \cdot)} = \{\Theta_{\alpha,k}(y, b) = \phi_\alpha(y)\psi_k(b)\}_{\alpha,k}$. The character $\chi \in \hat{G}$ is given by the pull-back through the isomorphism

$$\chi_{\alpha,k}(x, a) = \Theta_{\alpha,k}(\Psi(x, a)) = \Theta_{\alpha,k}(ax, a). \tag{2.34}$$

Therefore, we have $\chi_{\alpha,k}(x, a) = \phi_\alpha(ax)\psi_k(a) = \omega^{\mathrm{Tr}\langle \alpha, ax \rangle}\psi_k(a)$. □

2.4.2 Approximate representation and matrix lift of f

Definition 2.4.6 (Matrix lift F_f)**.** *Define a unitary matrix valued lift $F_f : G \to U(\mathbb{C}^{q-1})$ by the diagonal matrix*

$$F_f(x,a) = D(af(x)) := \sum_{c\in A} \omega^{\operatorname{Tr} caf(x)} |c\rangle\langle c|, \tag{2.35}$$

where $\{|c\rangle\}$ denotes the orthonormal basis vectors of a Hilbert space $\mathcal{H} \cong \mathbb{C}^{q-1}$ and the inner product between operators defined by $\langle A, B\rangle_\sigma = \operatorname{Tr}(A^* B\sigma)$ and choose $\sigma = \mathbb{I}/(q-1)$.

We first show that the BLR condition indeed gives an approximate represenatation.

Lemma 2.4.7 (Lift operation from BLR condition)**.** *Let $f : X \to \mathbb{F}_q$ and $(x,a),(y,b) \in G$ be rejected by the BLR test with probability ε, then the unitary matrix lift F_f in Equation 2.35 is an $(\frac{q}{q-1}\varepsilon, \frac{\mathbb{I}}{q-1})$ approximate representation of $(G,\star)$.*

Proof. The BLR test condition gives

$$\Pr_{c,d,x,y}\big[cf(x) + df(y) \neq f(cx+dy)\big] \leq \varepsilon.$$

First we observe that

$$\Pr_{g,h\in G}\big[F_f(g\star h) = F_f(g)F_f(h)\big] = \Pr_{c,d,x,y}\big[cf(x)+df(y) = f(cx+dy)\big]. \tag{2.36}$$

Using Equation 2.29 and Equation 2.35,

$$\begin{aligned} F_f\big((x,a)\star(y,b)\big) &= \sum_{c\in A} \omega^{\operatorname{Tr} cab\, f(b^{-1}x + a^{-1}y)} |c\rangle\langle c|, \\ F_f(x,a)\, F_f(y,b) &= \sum_{c\in A} \omega^{\operatorname{Tr} c(af(x)+bf(y))} |c\rangle\langle c|. \end{aligned}$$

These coincide iff their exponents agree modulo p for $c \in A$:

$$ab\, f(b^{-1}x + a^{-1}y) = af(x) + bf(y) \mod p, \quad \forall c \in A.$$

Therefore, we have equality over $\mathbb{F}_p$

$$ab\, f(b^{-1}x + a^{-1}y) = af(x) + bf(y).$$

Since a, b have a multiplicative inverse, substitute $c := b^{-1}$, $d := a^{-1}$, we have

$$f(cx+dy) = cf(x) + df(y),$$

which proves Equation 2.36.

To relate the test condition to the approximate representation, first note the equivalent definition of approximate representation

$$\mathbb{E}_{g,h\in G} \Re \left\langle F_f(g)^* F_f(h), F_f\left(g^{-1}h\right)\right\rangle_\sigma = \mathbb{E}_{g,h\in G} \Re \left\langle F_f(g) F_f(h), F_f\left(gh\right)\right\rangle_\sigma, \tag{2.37}$$

because F_f is unitary so $F_f(g)^* = F_f(g^{-1}), \forall g \in G$. Then evaluate the right hand side

$$\begin{aligned} \mathbb{E}_{g,h\in G} \Re \left\langle F_f(g)F_f(h), F_f(gh)\right\rangle_\sigma &= \mathbb{E}_{g,h\in G} \Re \operatorname{Tr}\left[F_f^*(h) F_f^*(g) F_f(gh)\,\sigma\right], \\ &= \mathbb{E}_{(x,a),(y,b)\in G} \frac{1}{q-1} \Re \operatorname{Tr}\left[\sum_{c\in A} \omega^{-\operatorname{Tr} c(af(x)+bf(y))} \omega^{\operatorname{Tr} cab\, f(b^{-1}x+a^{-1}y)} |c\rangle\langle c|\right], \\ &= \mathbb{E}_{(x,a),(y,b)\in G} \frac{1}{q-1} \Re \sum_{c\in A} \omega^{\operatorname{Tr} cz}, \end{aligned} \tag{2.38}$$

where we define $z = c(ab\, f(b^{-1}x + a^{-1}y) - (af(x) + bf(y)))$. To compute the expectation, we can split to two cases

$$\sum_{c\in A} \omega^{\mathrm{Tr}\, cz} = \begin{cases} q-1, & z = 0 \quad \text{when } f \text{ passes BLR,} \\ -1, & z \neq 0 \quad \text{when } f \text{ fails BLR.} \end{cases} \tag{2.39}$$

The second case is because $\sum_{c\in A} \omega^{\mathrm{Tr}\, cz} + 1 = \sum_{c\in \mathbb{F}_q} \omega^{\mathrm{Tr}\, cz} = 0$. Therefore, we have the approximate representation

$$\mathbb{E}_{g,h\in G} \Re \left\langle F_f(g) F_f(h), F_f(gh) \right\rangle_\sigma \geq (1-\varepsilon) - \frac{1}{q-1}\varepsilon = 1 - \frac{q}{q-1}\varepsilon. \tag{2.40}$$

Hence F_f is an approximate representation with $\epsilon = \frac{q}{q-1}\varepsilon$ as claimed.

□

The approximate representation "appears" slightly worse than what we would have hoped $\epsilon > \varepsilon$, but we will see this is not an issue for achieving a high soundness. Remarkably, the twisted group operation let us recover the linearity test condition exactly.

2.4.3 Gowers-Hatami implies BLR high soundness

In this section, we first prove the key lemma relating the implication of GH theorem to the hamming distance. The proof outlined here correspond to the formalized version in lean. The most convenient proof still uses properties of the Fourier transform of F_f. However, we remark that there are other alternative proofs, which we leave as comment in the current source file.

First let us write down statements about Fourier coefficients of F_f, which we will use in the proof.

Proposition 2.4.8 (Fourier coefficient of F_f). *Let the matrix lift $F_f : G \to U(\mathbb{C}^{q-1})$ be defined by the diagonal matrix*

$$F_f(x,a) = D(af(x)) := \sum_{c\in A} \omega^{\mathrm{Tr}\, caf(x)} |c\rangle\langle c|,$$

then its Fourier coefficients can be written as

$$[\hat{F}_f(\chi_{c\beta,k})]_{cc} = \psi_k(c)\nu_k(\beta), \tag{2.41}$$

where

$$\nu_k(\beta) := \mathbb{E}_{x,b} \left[\omega^{\mathrm{Tr} b(f(x) - \langle \beta, x\rangle)} \overline{\psi_k(b)}\right], \tag{2.42}$$

Proof. By Proposition 2.4.5, the characters are given by

$$\hat{G} = \{\chi_{\alpha,k}\}_{\alpha\in X, k\in[0,\cdots,q-2]}, \chi_{\alpha,k}(x,a) = \omega^{\mathrm{Tr}\langle \alpha, ax\rangle}\psi_k(a).$$

Therefore the Fourier coefficients of F_f are

$$\begin{aligned} \hat{F}_f(\chi_{\alpha,k}) &= \sum_{c\in \mathbb{F}_q^\times} \mathbb{E}_{x,a} \omega^{\mathrm{Tr}\, caf(x)} \bar{\chi}_{\alpha,k}(x,a), \\ &= \sum_{c\in \mathbb{F}_q^\times} \mathbb{E}_{x,a} \left[\omega^{\mathrm{Tr}\, a(cf(x) - \langle \alpha, x\rangle)} \overline{\psi_k(a)}\right] |c\rangle\langle c|. \end{aligned} \tag{2.43}$$

Since $c \in \mathbb{F}_q^\times$ is invertible, we perform a change of variables. Let $\alpha = c\beta$ for a fresh $\beta \in \mathbb{F}_q^n$. Because c is fixed, this is a bijection on $\mathbb{F}_q^n$. We also substitute $b = ca$, which is a bijection on $\mathbb{F}_q^\times$. This means $a = bc^{-1}$, and by the multiplicity of Dirichlet characters, $\overline{\psi_k(a)} = \overline{\psi_k(bc^{-1})} = \overline{\psi_k(b)}\psi_k(c)$.

Substituting these into the expectation isolates the c-dependence:

$$\begin{aligned}[\hat{F}_f(\chi_{c\beta,k})]_{cc} &= \mathbb{E}_{x,b}\left[\omega^{\mathrm{Tr} b(f(x)-\langle\beta,x\rangle)}\overline{\psi_k(b)}\psi_k(c)\right] \\ &= \psi_k(c)\nu_k(\beta),\end{aligned} \tag{2.44}$$

where $\nu_k(\beta) := \mathbb{E}_{x,b}\left[\omega^{\mathrm{Tr} b(f(x)-\langle\beta,x\rangle)}\overline{\psi_k(b)}\right]$ is a c-independent coefficient. $\square$

Lemma 2.4.9 (Gowers-Hatami to linearity (formalized proof))**.** *Suppose $F_f : G \to U(\mathbb{C}^{q-1})$ defined in Equation 2.35 via $f : \mathbb{F}_q^n \to \mathbb{F}_q$ is an (ϵ, σ) approximate unitary representation of $(G, \star)$ defined in Equation 2.29. Implicitly, $\epsilon \leq 1$.*

Then, there exists a linear function $\ell : \mathbb{F}_q^n \to \mathbb{F}_q$ such that the relative Hamming distance $\delta(f, \ell)$ is bounded by:

$$\delta(f, \ell) \leq \frac{q-1}{q}\epsilon. \tag{2.45}$$

Proof. Write the diagonal lift as $F_f(g) = \mathrm{diag}_{c\in\mathbb{F}_q^\times}(\varphi_c(g))$, where the scalar functions are $\varphi_c(x, a) = \omega^{\mathrm{Tr}(caf(x))}$.

Because F_f is diagonal and $\sigma = \mathbb{I}/(q-1)$, the trace equates to the average over the diagonal elements. By an intermediate statement in the proof of Gowers-Hatami theorem using Lemma. 2.3.2, the compression from an exact representation can be rewritten as a pair-correlation:

$$\mathcal{C}(f) := \mathbb{E}_{g,h}\Re\langle F_f(g)F_f(h), F_f(gh)\rangle_\sigma \geq 1 - \epsilon. \tag{2.46}$$

To connect the operator pair-correlation directly to the cubic trace of the Fourier coefficients, we express the lift matrices using the inverse Fourier transform, $F_f(x) = \sum_{\chi\in\widehat{G}} \hat{F}_f(\chi)\chi(x)$. Substituting this into the inner product definition of the pair-correlation and using the character homomorphism $\chi(gh) = \chi(g)\chi(h)$, we expand the trace:

$$\begin{aligned}\mathcal{C}(f) &= \mathbb{E}_{g,h\in G}\Re\mathrm{Tr}\left(F_f(g)^*F_f(h)^*F_f(gh)\sigma\right) \\ &= \Re\mathrm{Tr}\left(\sum_{\chi_1,\chi_2,\chi_3}\hat{F}_f^*(\chi_1)\hat{F}_f^*(\chi_2)\hat{F}_f(\chi_3)\sigma\cdot\mathbb{E}_g\left[\overline{\chi_1(g)}\chi_3(g)\right]\mathbb{E}_h\left[\overline{\chi_2(h)}\chi_3(h)\right]\right).\end{aligned} \tag{2.47}$$

By the orthogonality of characters, the expectations over g and h independently vanish unless $\chi_1 = \chi_3$ and $\chi_2 = \chi_3$. This completely collapses the triple sum to a single sum over χ. Because the Fourier coefficients of the diagonal lift F_f are themselves diagonal matrices (and thus commute), we can rearrange the surviving terms to recover the exact cubic trace expression:

$$\mathcal{C}(f) = \Re\sum_{\chi\in\widehat{G}}\mathrm{Tr}\left(\hat{F}_f^*(\chi)\hat{F}_f(\chi)\hat{F}_f^*(\chi)\sigma\right). \tag{2.48}$$

Because the Fourier coefficients are all fully diagonal since F_f is diagonal, we expand the product

$$\sum_{\chi_{\alpha,k}\in\widehat{G}}\mathrm{Tr}(\hat{F}_f^*(\chi)\hat{F}_f(\chi)\hat{F}_f^*(\chi)\sigma) = \frac{1}{q-1}\sum_{\alpha\in\mathbb{F}_q^n}\sum_{k=0}^{q-2}\sum_{c\in\mathbb{F}_q^\times}\left|[\hat{F}_f(\chi_{\alpha,k})]_{cc}\right|^2\overline{[\hat{F}_f(\chi_{\alpha,k})]_{cc}}.$$

Using the substitution $\alpha = c\beta$, summing over all α is equivalent to summing over all β. Substituting our factored coefficient from Eq. (2.41) in Proposition 2.4.8:

$$\begin{aligned}\sum_{\chi \in \widehat{G}} \mathrm{Tr}(\hat{F}_f^*(\chi)\hat{F}_f(\chi)\hat{F}_f^*(\chi)\sigma) &= \frac{1}{q-1}\sum_{c\in\mathbb{F}_q^\times}\sum_{\beta\in\mathbb{F}_q^n}\sum_{k=0}^{q-2}|\psi_k(c)\nu_k(\beta)|^2\overline{\psi_k(c)\nu_k(\beta)} \\ &= \frac{1}{q-1}\sum_{\beta,k}|\nu_k(\beta)|^2\overline{\nu_k(\beta)}\sum_{c\in\mathbb{F}_q^\times}\overline{\psi_k(c)},\end{aligned}$$

where

$$\nu_k(\beta) := \mathbb{E}_{x,b}\left[\omega^{b(f(x)-\langle\beta,x\rangle)}\overline{\psi_k(b)}\right],$$

is defined in Eq. (2.42).

Use the indicator function of $\mathbb{F}_q^\times$:

$$\sum_{c\in\mathbb{F}_q^\times}\overline{\psi_k(c)} = (q-1)\mathbb{I}(k=0)$$

we have

$$\sum_{\chi \in \widehat{G}} \mathrm{Tr}(\hat{F}_f^*(\chi)\hat{F}_f(\chi)\hat{F}_f^*(\chi)\sigma) = \sum_{\beta\in\mathbb{F}_q^n}|\nu_0(\beta)|^2\overline{\nu_0(\beta)}. \tag{2.49}$$

Note that $\nu_0(\beta) \in \mathbb{R}$ is related to the hamming distance $\delta(f,\ell_\beta)$ for liner function $\ell_\beta = \langle f, \ell_\beta\rangle$ via

$$\nu_0(\beta) = (1-\delta(f,\ell_\beta)) - \frac{1}{q-1}\delta(f,\ell_\beta) = 1 - \frac{q}{q-1}\delta(f,\ell_\beta). \tag{2.50}$$

Moreover, because the Fourier coefficients sum to identity by Eq. (**??**), we have

$$\begin{aligned}1 &= \frac{1}{q-1}\sum_{\chi\in\hat{G}}\mathrm{Tr}\left[\hat{F}_f^*(\chi)\hat{F}_f(\chi)\right], \\ &= \frac{1}{q-1}\sum_{c\in\mathbb{F}_q^\times}\sum_{\beta\in\mathbb{F}_q^n}\sum_k|\psi_k(c)\nu_k(\beta)|^2, \\ &= \sum_{\beta\in\mathbb{F}_q^n}\sum_k|\nu_k(\beta)|^2. \end{aligned} \tag{2.51}$$

Therefore $|\nu_k(\beta)|^2$ is a probability distribution indexed by (k,β). Because the $k=0$ probability is a subset, we have

$$\sum_{\beta\in\mathbb{F}_q^n}|\nu_0(\beta)|^2 \le \sum_{\beta\in\mathbb{F}_q^n}\sum_k|\nu_k(\beta)|^2 = 1. \tag{2.52}$$

Let β^* be the closest linear function such that

$$\nu_0(\beta^*) := \max_{\beta\in\mathbb{F}_q^n}\nu_0(\beta). \tag{2.53}$$

Therefore from Gowers-Hatami implication in Eq. (2.46) and the simplification of Fourier coefficeints in Eq. (2.49), we have the bound

$$1-\epsilon \le \Re\sum_{\chi\in\widehat{G}}\mathrm{Tr}(\hat{F}_f^*(\chi)\hat{F}_f(\chi)\hat{F}_f^*(\chi)\sigma) = \sum_{\beta\in\mathbb{F}_q^n}|\nu_0(\beta)|^2\overline{\nu_0(\beta)} \le \nu_0(\beta^*)\sum_{\beta\in\mathbb{F}_q^n}|\nu_0(\beta)|^2 \le \nu_0(\beta^*). \tag{2.54}$$

Unpack the relation to the hamming distance, we have

$$1 - \frac{q}{q-1}\delta(f, \ell_{\beta^*}) \geq 1 - \epsilon \implies \delta(f, \ell_{\beta^*}) \leq \frac{q-1}{q}\epsilon. \tag{2.55}$$

□

Chapter 3

Exponential-length PCP

Definition 3.0.1. *A system of m quadratic equations in n variables over a field $\mathbb{F}$ is a list of polynomials $p_1, \dots, p_m \in \mathbb{F}[x_1, \dots, x_n]$ where each p_i has total degree at most 2.*

$$\mathrm{QESAT}(\mathbb{F}, n) := \{(p_1, \dots, p_m) \mid \exists a_1, \dots, a_n \in \mathbb{F},\ \forall i \in [m],\ p_i(a_1, \dots, a_n) = 0\}.$$

For example, $(x + 1, xy + z) \in \mathrm{QESAT}(\mathbb{F}_2, 3)$.

Definition 3.0.2. *A* PCP verifier *is a probabilistic oracle Turing machine V with query access to a function $\pi : [\ell(|x|)] \to \Sigma$.*

Definition 3.0.3. *A language $L \subseteq \{0,1\}^*$ is in $\mathrm{PCP}[\varepsilon_c, \varepsilon_s, \Sigma, \ell, q, r]$ if there exists a polynomial-time PCP verifier V such that for every $x \in \{0,1\}^*$, V makes at most $q(|x|)$ queries to the proof oracle, uses at most $r(|x|)$ bits of randomness, and the following holds:*

- *Completeness: If $x \in L$, then $\exists \pi \in \Sigma^{\ell(|x|)}$, $\Pr\left[V^{\pi}(x) = 1\right] \geq 1 - \varepsilon_c$.*
- *Soundness: If $x \notin L$, then $\forall \tilde{\pi} \in \Sigma^{\ell(|x|)}$, $\Pr\left[V^{\tilde{\pi}}(x) = 1\right] \leq \varepsilon_s$.*

The following theorem is the main goal of this chapter.

Theorem 3.0.4.

$$\mathrm{QESAT}(\mathbb{F}_2, n) \in \mathrm{PCP}[\varepsilon_c = 0, \varepsilon_s = 1/2, \Sigma = \mathbb{F}_2, \ell = 2^{|x|}, q = O(1), r = |x|^{O(1)}].$$

Proof. TODO ◻

Definition 3.0.5. *A* LPCP verifier *is a probabilistic oracle Turing machine V with query access to a linear function $\langle \pi, \cdot \rangle : \Sigma^{\ell(|x|)} \to \Sigma$.*

Definition 3.0.6. *A language $L \subseteq \{0,1\}^*$ is in $\mathrm{LPCP}[\varepsilon_c, \varepsilon_s, \Sigma, \ell, q, r]$ if there exists a polynomial-time LPCP verifier V such that for every $x \in \{0,1\}^*$, V makes at most $q(|x|)$ queries to the linear proof oracle, uses at most $r(|x|)$ bits of randomness, and the following holds:*

- *Completeness: If $x \in L$, then $\exists \pi \in \Sigma^{\ell(|x|)}$, $\Pr\left[V^{\langle \pi, \cdot \rangle}(x) = 1\right] \geq 1 - \varepsilon_c$.*
- *Soundness: If $x \notin L$, then $\forall \tilde{\pi} \in \Sigma^{\ell(|x|)}$, $\Pr\left[V^{\langle \tilde{\pi}, \cdot \rangle}(x) = 1\right] \leq \varepsilon_s$.*

3.1 Simpler BLR verifier for $\mathbb{F}_2$

Definition 3.1.1. *Let $V^f_{\mathrm{BLR}(\mathbb{F}_2,n)}$ be the verifier with oracle access to $f : \mathbb{F}_2^n \to \mathbb{F}_2$ defined as follows:*

1. *Sample $x, y \leftarrow \mathbb{F}^n$.*
2. *Accept if and only if $f(x) + f(y) = f(x + y)$.*

Theorem 3.1.2. *Let $n > 0$. For every linear function $f : \mathbb{F}_2^n \to \mathbb{F}_2$, we have*

$$\Pr\left[V^f_{\mathrm{BLR}(\mathbb{F}_2,n)} = 1\right] = 1.$$

Proof. □

Theorem 3.1.3. *Let $n > 0$ and $f : \mathbb{F}_2^n \to \mathbb{F}_2$.*

$$\Pr\left[V^f_{\mathrm{BLR}(\mathbb{F}_2,n)} = 0\right] \geq \Delta(f, \mathcal{LIN}).$$

Proof. □

3.2 Linear PCP for linear equations

Definition 3.2.1. *For a field $\mathbb{F}$ and parameters $m, n \in \mathbb{N}$, we define the language*

$$\mathrm{LINEQ}(\mathbb{F}, m, n) := \{(M, c) \in \mathbb{F}^{m\times n} \times \mathbb{F}^m \mid \exists b \in \mathbb{F}^n,\ Mb = c\}.$$

Definition 3.2.2. *Let $V^{\langle b,\cdot\rangle}_{\mathrm{LINEQ}(\mathbb{F},m,n)}(M, c)$ be the LPCP verifier defined as follows:*

1. *Sample $r \leftarrow \mathbb{F}^m$.*
2. *Query $b \in \mathbb{F}^n$ at $u := M^\top r \in \mathbb{F}^n$.*
3. *Accept if and only if $\langle b, u\rangle = \langle c, r\rangle$.*

Theorem 3.2.3.

$$\mathrm{LINEQ}(\mathbb{F}, m, n) \in \mathrm{LPCP}\left[\varepsilon_c = 0, \varepsilon_s = 1/|\mathbb{F}|, \Sigma = \mathbb{F}, \ell = |x|, q = 1, r = m \log |\mathbb{F}|\right].$$

Proof. Consider the LPCP verifier $V^{\langle b,\cdot\rangle}_{\mathrm{LINEQ}(\mathbb{F},m,n)}(M, c)$.
Completeness: If $Mb = c$, then for every $r \in \mathbb{F}^m$, we have

$$\langle b, u\rangle = b^\top(M^\top r) = (Mb)^\top r = \langle c, r\rangle.$$

Soundness: If $Mb \neq c$, then

$$\begin{aligned}\Pr_{r\leftarrow\mathbb{F}^m}\left[\langle b, u\rangle = \langle c, r\rangle\right] &= \Pr_{r\leftarrow\mathbb{F}^m}\left[\langle Mb, r\rangle = \langle c, r\rangle\right] \\ &= \Pr_{r\leftarrow\mathbb{F}^m}\left[\sum_{i=1}^{m}(Mb - c)_i \cdot r_i = 0\right] \leq 1/|\mathbb{F}|,\end{aligned}$$

where the last inequality follows from the Schwartz-Zippel lemma. □

3.3 Linear PCP for tensor checks

Definition 3.3.1. *For a field $\mathbb{F}$, vectors $a \in \mathbb{F}^n$ and $b \in \mathbb{F}^m$, the tensor product $a \otimes b \in \mathbb{F}^{n \times m}$ is the matrix defined by*

$$(a \otimes b)_{i,j} := a_i \cdot b_j \quad \forall i \in [n],\, j \in [m].$$

We write $\mathrm{flat}(a \otimes b) \in \mathbb{F}^{n \cdot m}$ *for the row-concatenation of $a \otimes b$.*

For our purposes, we would love to test to check if $\mathrm{flat}(a \otimes a) = b$

Definition 3.3.2. *For a field $\mathbb{F}$ and $n \in \mathbb{N}$, define the tensor test language*

$$\mathrm{TENSORQ}(\mathbb{F}, n) := \{(a, b) \in \mathbb{F}^n \times \mathbb{F}^{n^2} \mid b = \mathrm{flat}(a \otimes a)\}.$$

Theorem 3.3.3. *For every finite field $\mathbb{F}$ and $n \in \mathbb{N}$,*

$$\mathrm{TENSORQ}(\mathbb{F}, n) \in \mathrm{LPCP}\left[\varepsilon_c = 0,\ \varepsilon_s = \frac{2|\mathbb{F}| - 1}{|\mathbb{F}|^2},\ \Sigma = \mathbb{F},\ \ell = n + n^2,\ q = 3,\ r = 2n \cdot \log(|\mathbb{F}|)\right].$$

Proof. Consider the following LPCP verifier $V^{\langle a, \cdot \rangle, \langle b, \cdot \rangle}$ proceeds as follows:

1. Sample $s, t \leftarrow \mathbb{F}^n$.
2. Query a at s and at t, obtaining $\langle a, s \rangle$ and $\langle a, t \rangle$.
3. Query b at $\mathrm{flat}(s \otimes t)$, obtaining $\langle b, \mathrm{flat}(s \otimes t) \rangle$.
4. Check if $\langle b, \mathrm{flat}(s \otimes t) \rangle = \langle a, s \rangle \cdot \langle a, t \rangle$.

Completeness. Suppose $b = \mathrm{flat}(a \otimes a)$. Then $\forall s, t \in \mathbb{F}^n$,

$$\begin{aligned} \langle b,\, \mathrm{flat}(s \otimes t) \rangle = \langle \mathrm{flat}(a \otimes a),\, \mathrm{flat}(s \otimes t) \rangle &= \sum_{i,j \in [n]} a_i a_j s_i t_j \\ &= \Big(\sum_{i \in [n]} a_i s_i\Big) \cdot \Big(\sum_{j \in [n]} a_j t_j\Big) = \langle a, s \rangle \cdot \langle a, t \rangle, \end{aligned}$$

Soundness. Suppose $b \neq \mathrm{flat}(a \otimes a)$, i.e., there exists $i^*, j^* \in [n]$ such that $b_{i^* j^*} \neq a_{i^*} \cdot a_{j^*}$. For each $i \in [n]$ define the linear polynomial $p_i(t) := \sum_{j \in [n]} (b_{ij} - a_i a_j)\, t_j$. The check can be rewritten as

$$\langle b, \mathrm{flat}(s \otimes t) \rangle - \langle a, s \rangle \langle a, t \rangle = \sum_{i \in [n]} \sum_{j \in [n]} (b_{ij} - a_i a_j)\, s_i t_j = \sum_{i \in [n]} p_i(t)\, s_i.$$

Immediate application of Polynomial Identity Testing yields a lower bound of $1 - \frac{2}{|\mathbb{F}|}$ but we can do better. Since $b_{i^* j^*} \neq a_{i^*} a_{j^*}$, the polynomial p_{i^*} is nonzero, so by the Schwartz–Zippel lemma applied to t,

$$\Pr_t[p_{i^*}(t) \neq 0] \geq 1 - \tfrac{1}{|\mathbb{F}|}.$$

Conditioned on $p_{i^*}(t) \neq 0$, the expression $\sum_i p_i(t)\, s_i$ is a nonzero linear polynomial in s, so by the Schwartz–Zippel lemma applied to s,

$$\Pr_{s \leftarrow \mathbb{F}^n}\Big[\sum_i p_i(t)\, s_i \neq 0 \,\Big|\, p_{i^*}(t) \neq 0\Big] \geq 1 - \tfrac{1}{|\mathbb{F}|}.$$

Hence

$$\Pr_{s,t}[\text{verifier rejects}] \;\geq\; \left(1 - \tfrac{1}{|\mathbb{F}|}\right)^2,$$

and therefore

$$\Pr_{s,t}[\text{verifier accepts}] \;\leq\; 1 - \left(1 - \tfrac{1}{|\mathbb{F}|}\right)^2 = \frac{2|\mathbb{F}| - 1}{|\mathbb{F}|^2}. \qquad \square$$

Corollary 3.3.4.

$$\mathrm{TENSORQ}(\mathbb{F}_2, n) \in \mathrm{LPCP}\left[\varepsilon_c = 0,\ \varepsilon_s = \frac{3}{4},\ \Sigma = \mathbb{F}_2,\ \ell = n + n^2,\ q = 3,\ r = 2 \cdot n\right].$$

Proof. Immediate. □

3.4 Linear PCP to PCP

Lemma 3.4.1 (Chernoff bound)**.** *Let $X_1, \ldots, X_N$ be independent random variables over $\{0,1\}$, and let $\mu = \mathbb{E}\left[\sum_{i=1}^N X_i\right]$. For any $\Delta > 0$ one has*

$$\Pr\left[\sum_{i=1}^{N} X_i \geq \mu + \Delta\right] \leq \exp(-2\Delta^2/N)$$

Proof. Write $p_i = \mathbb{E}[X_i]$, so that $\mu = \sum_{i=1}^N p_i$. We first prove the elementary exponential-moment estimate

$$\mathbb{E}[\exp(\lambda(X_i - p_i))] \leq \exp(\lambda^2/8)$$

for every $\lambda > 0$ and every i. Since X_i is $\{0,1\}$-valued,

$$\mathbb{E}[\exp(\lambda(X_i - p_i))] = (1 - p_i)\exp(-\lambda p_i) + p_i \exp(\lambda(1 - p_i)).$$

Equivalently, it is enough to show

$$\log(1 - p_i + p_i \exp(\lambda)) - p_i\lambda \leq \lambda^2/8.$$

For $p \in [0,1]$, set $f(\lambda) = \log(1 - p + p\exp(\lambda)) - p\lambda$. Then $f(0) = f'(0) = 0$, and

$$f''(\lambda) = \frac{p\exp(\lambda)}{1 - p + p\exp(\lambda)}\left(1 - \frac{p\exp(\lambda)}{1 - p + p\exp(\lambda)}\right) \leq \frac{1}{4}.$$

Hence, for $\lambda > 0$,

$$f(\lambda) = \int_0^{\lambda}\int_0^{u} f''(v)\,dv\,du \leq \int_0^{\lambda}\int_0^{u} \frac{1}{4}\,dv\,du = \lambda^2/8.$$

Let $S = \sum_{i=1}^N X_i$. By independence and the estimate above,

$$\mathbb{E}[\exp(\lambda(S - \mu))] = \prod_{i=1}^{N} \mathbb{E}[\exp(\lambda(X_i - p_i))] \leq \exp(N\lambda^2/8).$$

Markov's inequality gives, for every $\lambda > 0$,

$$\Pr[S \geq \mu + \Delta] = \Pr[\exp(\lambda(S - \mu)) \geq \exp(\lambda\Delta)] \leq \exp(-\lambda\Delta)\mathbb{E}[\exp(\lambda(S - \mu))]$$

Using the moment bound above, this is at most

$$\exp(-\lambda\Delta + N\lambda^2/8).$$

Taking $\lambda = 4\Delta/N$ gives

$$\Pr[S \geq \mu + \Delta] \leq \exp(-2\Delta^2/N).$$

□

Lemma 3.4.2.

$\mathrm{LPCP}[\varepsilon_c, \varepsilon_s, \Sigma = \mathbb{F}_2, \ell, q, r] \subseteq \mathrm{PCP}\,[\varepsilon_c,\ \varepsilon_s' = \max\{7/8, \varepsilon_s + 1/100\}, \Sigma' = \mathbb{F}_2,\ \ell' = 2^\ell,\ q' = 3 + 16\lceil\log(100q)\rceil q,\ r' =$

Proof. Let $L \in \mathrm{LPCP}[\varepsilon_c, \varepsilon_s, \Sigma = \mathbb{F}_2, \ell, q, r]$, and let V be the LPCP verifier for L. Set $t = 8\lceil\log(100q)\rceil$, where log denotes the natural logarithm. We define a PCP verifier as follows, for any $\bar{pi} \in \mathbb{F}_2^{\mathbb{F}_2^\ell}$ and $x \in \{0,1\}^*$.

Verifier $\bar{V}^{\bar{\pi}}(x)$:

1. If $V_{\mathrm{BLR}}^{\bar{\pi}} = 0$ output 0 and halt, otherwise proceed with the next step.
2. Independently for each $i \in [t]$, sample $r_i \leftarrow \mathbb{F}_2^\ell$.
3. Simulate V by answering each linear query a with $\mathrm{plurality}(\bar{\pi}(a + r_i) - \bar{\pi}(r_i))_{i\in[t]}$.

Claim 3.4.3 (Completeness)**.** *For every $x \in L$, there exists $\bar{\pi} \in \mathbb{F}_2^{\mathbb{F}_2^\ell}$ such that* $\Pr[\bar{V}^{\bar{\pi}}(x) = 1] \geq 1 - \varepsilon_c$.

Proof. Since $x \in L$ and $L \in \mathrm{LPCP}[\varepsilon_c, \varepsilon_s, \Sigma = \mathbb{F}_2, \ell, q, r]$, let $\pi \in \mathbb{F}_2^\ell$ such that $\Pr[V^{\langle\pi,\cdot\rangle}(x) = 1] \geq 1 - \varepsilon_c$. Then we construct $\bar{\pi} \in \mathbb{F}_2^{\mathbb{F}_2^\ell}$ as follows: we define $\bar{\pi}(x) = \langle\pi, x\rangle$ for each $x \in \mathbb{F}^\ell$. Using the completeness of V_{BLR} we know that $V_{\mathrm{BLR}}^{\bar{\pi}} = 1$ with probability 1. Also, $\mathrm{plurality}(\bar{\pi}(a+r_i) - \bar{\pi}(r_i))_{i\in[t]} = \langle\pi, a\rangle$ for any query a and any r_i. We then conclude that $V^{\langle\pi,\cdot\rangle}(x) = \bar{V}^{\bar{\pi}}(x)$ for any $r_1, \dots, r_t$. Thus $\Pr[\bar{V}^{\bar{\pi}}(x) = 1] = \Pr[V^{\langle\pi,\cdot\rangle}(x) = 1] \geq 1 - \varepsilon_c$. □

Claim 3.4.4 (Soundness)**.** *For every $x \notin L$ and every $\hat{\pi} \in \mathbb{F}_2^{\mathbb{F}_2^\ell}$, one has* $\Pr[\bar{V}^{\hat{\pi}}(x) = 1] \leq \max\{7/8, \varepsilon_s + 1/100\}$.

Proof. Let $x \notin L$ and $\hat{\pi} \in \mathbb{F}_2^{\mathbb{F}_2^\ell}$. We consider two cases.

Case $\delta(\hat{\pi}, \mathbf{LIN}) \geq 1/8$. Using the soundness of V_{BLR}, we know that $\Pr[V_{\mathrm{BLR}}^{\bar{\pi}} = 0] \geq 1/8$. Since $\Pr[\bar{V}^{\hat{\pi}}(x) = 1] \leq \Pr[V_{\mathrm{BLR}}^{\bar{\pi}} = 1]$ we conclude $\Pr[\bar{V}^{\hat{\pi}}(x) = 1] \leq 1 - 1/8 = 7/8$.

Case $\delta(\hat{\pi}, \mathbf{LIN}) < 1/8$ Let $S = \{\tilde{\pi} \in \mathcal{LIN} : \delta(\hat{\pi}, \tilde{\pi} \leq \delta(\hat{\pi}, \mathrm{LIN}))\}$. Suppose for a contradiction that $|S| \geq 2$, and let $\bar{\pi}, \tilde{\pi} \in S$ be distinct: since $\delta(\hat{\pi}, \bar{\pi} \leq \delta(\hat{\pi}, \mathrm{LIN})) < 1/8$ and $\delta(\hat{\pi}, \tilde{\pi} \leq \delta(\hat{\pi}, \mathrm{LIN})) < 1/8$, by triangle inequality we have $\delta(\bar{\pi}, \tilde{\pi}) < 1/4$; since $|\mathbb{F}_2| = 2$, this contradicts the fact that any two distinct $\bar{\pi}, \tilde{\pi} \in \mathcal{LIN}$ satisfy $\delta(\bar{\pi}, \tilde{\pi}) \geq 1/2$.

Let then $\tilde{\pi}$ be the unique element of $\mathcal{LIN}$ with minimal distance to $\hat{\pi}$, and let $z \in \mathbb{F}^\ell$ the unique vector such that $\tilde{\pi}(x) = \langle z, x\rangle$ for all x. If for every query a that V asks we have $\mathrm{plurality}(\hat{\pi}(a+r_i) - \hat{\pi}(r_i))_{i\in[t]} = \tilde{\pi}(a)$, then $\bar{V}^{\hat{\pi}}(x) = V^{\langle z,\cdot\rangle}(x)$, so provided that the r_i satisfy this condition we have that $\Pr[\bar{V}^{\hat{\pi}}(x) = 1] = \Pr[V^{\langle z,\cdot\rangle}(x) = 1] \leq \varepsilon_s$. We are only left to bound the probability that for each of the q queries that V asks we have $\mathrm{plurality}(\hat{\pi}(a+r_i) - \hat{\pi}(r_i))_{i\in[t]} = \tilde{\pi}(a)$.

Note that for any a and r_i, if $\hat{\pi}(a+r_i) = \tilde{\pi}(a+r_i)$ and $\hat{\pi}(r_i) = \tilde{\pi}(r_i)$ then $\hat{\pi}(a+r_i) - \hat{\pi}(r_i) = \tilde{\pi}(a)$. Hence, the proabability over one r_i that $\hat{\pi}(a+r_i) - \hat{\pi}(r_i) \neq \tilde{\pi}(a)$ is at most $2 \cdot \delta(\hat{\pi}, \tilde{\pi}) \leq 1/4$ by a union bound. Also note that plurality$(\hat{\pi}(a+r_i) - \hat{\pi}(r_i))_{i \in [t]} \neq \tilde{\pi}(a)$ means that $\hat{\pi}(a+r_i) - \hat{\pi}(r_i) \neq \tilde{\pi}(a)$ for at least $\lceil t/2 \rceil \geq t/2$ of the i's. Let $X_i = \mathbb{1}[\hat{\pi}(a+r_i) - \hat{\pi}(r_i) \neq \tilde{\pi}(a)]$. Then $\mu = \mathbb{E}[\sum_{i \in [t]} X_i] \leq t/4$. Applying Chernoff bound with $\Delta = t/4$, the probability over the choice of $r_1, \ldots, r_t$ that there are at least $t/2$ i's with $X_i = 1$ is at most $\exp(-t/8)$. Union bounding over the q queries that V asks, we have that the local correction fails with probability at most $q \cdot \exp(-t/8) \leq 1/100$ by our choice of t. □

□

Theorem 3.4.5.

$$\mathrm{QESAT}(\mathbb{F}_2, n) \in \mathrm{LPCP}[\varepsilon_c = 0, \varepsilon_s = 3/4, \Sigma = \mathbb{F}_2, \ell = n + n^2, q = 4, r = |x| + 2 * n].$$

Proof. Let $(p_1, \ldots, p_m)$ be an instance of QESAT($\mathbb{F}$) with n variables.
The LPCP verifier expects a proof $\pi = (a, b) \in \mathbb{F}^{n+n^2}$ and works as follows:

Verifier $V^{(a,b)}(p_1, \ldots, p_m)$:

1. Sample $r \leftarrow \mathbb{F}^m$ and $s, t \leftarrow \mathbb{F}^n$.
2. Define
$$M := \begin{bmatrix} \mathrm{coeff}(p_1) \\ \vdots \\ \mathrm{coeff}(p_m) \end{bmatrix} \in \mathbb{F}^{m \times (n+n^2)} \quad \text{and} \quad c := \begin{bmatrix} -\mathrm{const}(p_1) \\ \vdots \\ -\mathrm{const}(p_m) \end{bmatrix} \in \mathbb{F}^m.$$
(Note: coeff(p_i) := non-constant coeffs of p_i, and const(p_i) := constant coeff of p_i)
3. *(linear equation test)* → Query (a, b) at $M^T r$ and check that $\langle a \parallel b, M^T r \rangle = \langle c, r \rangle$.
4. *(tensor test)* → Query b at $\mathrm{flat}(s \otimes t)$, a at s and t, and check that
$$\langle b, \mathrm{flat}(s \otimes t) \rangle = \langle a, s \rangle \cdot \langle a, t \rangle.$$

Completeness: Suppose $p_1(a) = \cdots = p_m(a) = 0$ and set $b := \mathrm{flat}(a \otimes a)$.
Then
$$\Pr_{s,t}\left[\langle b, \mathrm{flat}(s \otimes t) \rangle = \langle a, s \rangle \cdot \langle a, t \rangle\right] = 1$$
and
$$\Pr_{r}\left[\langle a \parallel b, M^T r \rangle = \langle c, r \rangle\right] = 1 \qquad \left(\text{since } M \begin{bmatrix} a \\ \mathrm{flat}(a \otimes a) \end{bmatrix} = c\right).$$

Soundness: Suppose $\forall a \in \mathbb{F}^n \; \exists i \in [m]$ s.t. $p_i(a) \neq 0$. Fix any $\pi = (a, b)$.
Either:

(i) $b \neq \mathrm{flat}(a \otimes a) \implies$ tensor test passes w.p. $\leq \frac{2|\mathbb{F}|-1}{|\mathbb{F}|^2}$

OR (ii) $b = \mathrm{flat}(a \otimes a)$ and $M \begin{bmatrix} a \\ b \end{bmatrix} \neq c \implies$ linear equation test passes w.p. $\leq \frac{1}{|\mathbb{F}|}$

□